

Which statistical hypotheses are afflicted by false confidence?

Ryan Martin¹
North Carolina State University
`www4.stat.ncsu.edu/~rgmarti3`

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Statisticians want numerical measures of the degree to which data support hypotheses. —Hacking

Even if a frequency-based view of probability is not used directly as a basis for inference, it is unacceptable if a procedure... of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions. —Reid & Cox

Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information. —Efron

Statistical Science
2011, Vol. 26, No. 3, 299–316
DOI: 10.1214/11-STS352
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Is Bayes Posterior just Quick and Dirty Confidence?¹

D. A. S. Fraser

- Above quotes seem to point to a “reliable Bayesian UQ”²
- However: Bayes doesn’t reliably quantify uncertainty
- *Imprecise probabilities are necessary!*
- Relevant questions:
 - mathematically describe this lack of reliability? ✓
 - develop an alternative framework for reliable UQ? ✓
 - ...
 - exactly diagnose the lack of reliability? □

²This includes (generalized) fiducial and other variants

- Data X is informative about uncertain Θ
- Connection is via a statistical model: $(X \mid \Theta = \theta) \sim P_\theta$
- Without much³ loss of generality, I'll focus mainly on

$$P_\theta = N_D(\underbrace{\theta \in \mathbb{R}^D}_{\text{uncertain}}, \underbrace{\Sigma \in \mathbb{R}_{\text{pd}}^{D \times D}}_{\text{certain}})$$

- No nuisance parameters, prior info is vacuous
- Let Q_x be a “posterior” for Θ , given $X = x$
 - default-prior Bayes
 - (generalized) fiducial
 - ...
- Normal model: $Q_x = N_D(x, \Sigma)$

³See M. and Williams (*BELIEF*'24)

- Natural interpretation of Q_x , for $X = x$:
 - $Q_x(H)$ large $\implies$ infer " $\Theta \in H$ " (confidently)
 - $Q_x(H)$ small $\implies$ infer " $\Theta \notin H$ " (confidently)
- Then the following would be problematic:
 - for some false hypotheses $H \not\supseteq \Theta$
 - the rv $X \mapsto Q_X(H)$ tends to be large under P_Θ
- *False confidence*⁴ is a particular brand of unreliability, and all probabilistic UQ is afflicted by it

False confidence theorem.

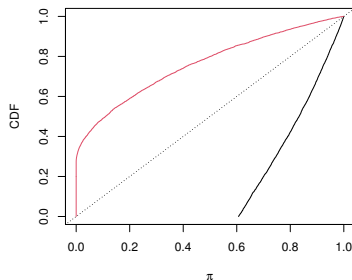
Let Q_X be *any* data-dependent probability for Θ . For any thresholds (τ, ρ) , there exists hypotheses $H \subset \mathbb{T}$ such that

$$H \not\supseteq \Theta \quad \text{and} \quad P_\Theta\{Q_X(H) > \tau\} > \rho.$$

⁴Balch, M., & Ferson 2019, *Proc Roy Soc A*, arXiv:1706.08565

False confidence, cont.

- Gaussian case, $D = 2$
 - $H = \{\theta : \|\theta\| > \|\Theta\|\}$, false!
 - $Q_X(H)$ is a non-central chi-square calculation
- Plot shows CDF $\pi \mapsto P_{\Theta}\{Q_X(H) \leq \pi\}$, with $\|\Theta\| = 1$
- Bayes assigns probability at least 0.6 to false H !



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Research



Satellite conjunction analysis and the false confidence theorem

Michael Scott Balch¹, Ryan Martin² and Scott Ferson³



International Journal of Approximate Reasoning

Volume 113, October 2019, Pages 39-73



False confidence, non-additive beliefs, and valid statistical inference

Ryan Martin ✉

Avoiding false confidence

- The false confidence theorem says “any probability Q_X ”
- What I mean is “any precise probability Q_X ”
- Valid inference, free of false confidence, can be achieved with a suitable data-dependent imprecise probability
- My suggested approach⁵ is based on “ Q_X ” taking the form of a consonant belief function

Valid and efficient imprecise-probabilistic inference
with partial priors, II. General framework

Ryan Martin*

⁵See, e.g., my talk at the BFAS School'23

Diagnosing false confidence

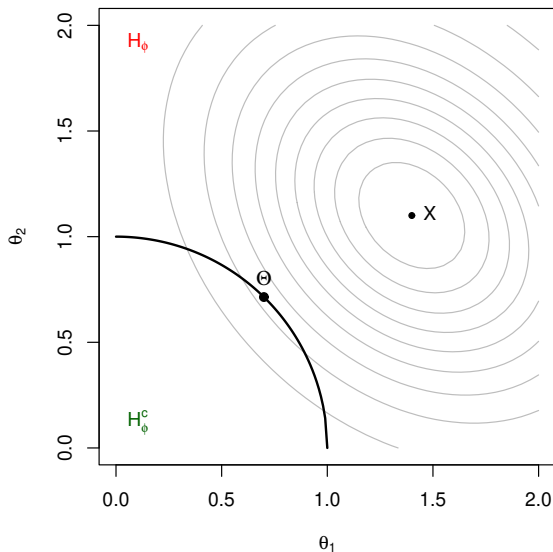
- False confidence theorem is an *existence result*
- Not a practically forceful theorem if the H 's afflicted with false confidence are rare/trivial/etc
- **Question:** Characterize H 's afflicted with false confidence?
- My/our conjecture is that *non-linear hypotheses* H about Θ are afflicted, but not sure how to prove it
- Results in the proceedings paper⁶ show progress
 - focus on the special normal model
 - identify a class of non-linear H 's that are afflicted
 - not a complete characterization (yet)

⁶Updates will probably be posted at [arXiv:2404.16228](https://arxiv.org/abs/2404.16228)

Diagnosing false confidence, cont.

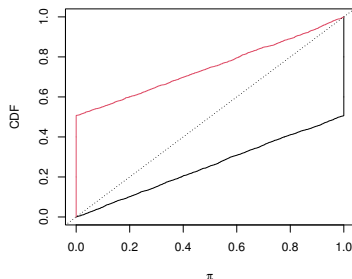
- Intuition in a stylized $Q_X = N_D(X, \Sigma)$ case
- Let $\phi : \mathbb{R}^D \rightarrow \mathbb{R}$ be a convex function
- False hypothesis $H_\phi = \{\theta : \phi(\theta) > \phi(\Theta)\}$
- Jensen's inequality: $E_\Theta\{\phi(X)\} > \phi(E_\Theta X) = \phi(\Theta)$
- Consequences:
 - posterior mode, X , tends to be in the false H_ϕ
 - contours of Q_X tend to be oriented away from true H_ϕ^c
- Now it's pretty easy to see why Q_X tends to assign high probability to false H_ϕ — a picture will help
- Paper fills in the (boring) details

Diagnosing false confidence, cont.



Example

- False confidence can appear due to “hidden” non-linearity
- Gaussian case, $D = 1$
 - non-trivial parameter constraint, e.g., $\mathbb{T} = [0, \infty)$
 - false hypothesis $H = (\Theta, \infty)$
- Plot shows CDF $\pi \mapsto P_{\Theta}\{Q_X(H) \leq \pi\}$, with $\Theta = 1$
- Bayes assigns probability 1 to false H 50% of the time!



[Bayes's formula] does not create real probabilities from hypothetical probabilities. —Fraser

- Bayes and its variants are afflicted with false confidence
- Now we know the extent of affliction a little better⁷
- I didn't say anything about imprecise prob/belief functions, but the results are still relevant to your community
 - false confidence = risk
 - if people understand risk, then they'll try to avoid it
 - the only way to avoid the is via imprecise UQ
 - that's what y'all are experts in!

⁷Paper's results are bit more general than I said here...



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

`https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/`

- Links to papers, talks, etc. can be found on my website:

`www4.stat.ncsu.edu/~rmartin/`

- If you have questions, etc, email me at `rgmarti3@ncsu.edu`

Thanks for your attention!